Zhenyu Wu

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Research Interests

My research interests include **Robot Learning**, **Humanoid Loco-manipulation**, and **Dexterous Manipulation**. I aim to develop **learned whole-body controllers** that are expressive, robust, and capable of contact-rich interaction, and to study how to integrate them with **high-level reasoning** for autonomous, general-purpose robots in the real world.

Education

University of California San Diego, Master of Science in Computer Science — Sept 2024 – Jun 2026

- GPA: 3.87/4.0

Lehigh University, Bachelor of Science in Computer Science — Aug 2019 – May 2024

- GPA: 3.82/4.0, Graduated with Highest Honors

Publications

[1] **TAPNAV: Humanoid Navigation through Tactile Active Perception** — Sep 2026
Zhenyu Wu[*], Huaze Liu[*], Jaehwi Jang[*], Junjie Sheng, Andrew Collins, Aaron Xie, Zhaoyuan Gu, Kaijie Zhu, Ding Jiang, Kevin Cai, Ye Zhao
Under review
https://tapnav.github.io/TAPNAV-Project/

[2] **Dataset-Free Compliant Humanoid Loco-Manipulation with Dynamic Online Posture** — Sep 2026
Seungho Yeom, *Zhenyu Wu*[†], Jaeyoung Huh[†], Diego Williams[†], Yuheng Zhi, Soofiyan Atar, Michael Yip
Under review | *CoRL 2026 Workshop on Open Problems in Contact-Rich Loco-Manipulation (Spotlight)*
Best Paper Award, IROS 2026 Workshop on *Bridging Learning and Physics for Robust Whole-Body Robot Control*
https://oclo-humanoid.github.io/

[3] **SteadyTray: Learning Object Balancing Tasks in Humanoid Tray Transport via Residual Reinforcement Learning** — Mar 2026
Zhenyu Wu[*], Anlun Huang[*], Soofiyan Atar[†], Yuheng Zhi[†], Michael Yip
2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, PA, USA
Winner, IROS Best Paper Award on Mobile Manipulation | Finalist, IROS Best Application Paper Award
https://steadytray.github.io/

[4] **Methods for Setting Device Specifications for Analog In-Memory Computing Inference** — Mar 2026
Zhenyu Wu, Xin Su, Malte J. Rasch, Ning Li
Advanced Intelligent Systems, vol. 8, no. 5, 2026

[*] / [†] Equal contribution

Research Experience

Research Assistant at Georgia Institute of Technology – Atlanta, GA — Jul 2026 – Present

Laboratory for Intelligent Decision and Autonomous Robots (LIDAR), Advised by Prof. Ye Zhao

TAPNAV: Humanoid Navigation through Tactile Active Perception

- Introduced a **humanoid tactile navigation** framework for vision-denied environments, where the robot localizes by actively probing nearby structures.
- Developed a **whole-body control policy** tracking end-effector poses, contact forces, and footsteps, trained with offline IK data, a virtual spring-damper field, and 3D-LIPM references.
- Built a **ROS 2 full-stack pipeline** integrating tactile sensing, proprioception, control, and global/local planning.

Graduate Research Assistant at University of California San Diego – San Diego, CA Sept 2024 – Jul 2026

Advanced Robotics and Controls Laboratory (ARClab), Advised by Prof. Michael Yip

SteadyTray: Learning Object Balancing Tasks in Humanoid Tray Transport via Residual Reinforcement Learning

- Introduced **ReST-RL**, a student-teacher residual RL framework that keeps unsecured objects on a tray stable while the humanoid walks and **actively rebalances them under disturbances**.
- Trained a **residual module** on top of a base walking policy in Isaac Lab, **distilling** a teacher with privileged object states into a student that sees only the AprilTag pose.
- Achieved **zero-shot sim-to-real** transfer on a Unitree G1, carrying diverse objects such as a water-filled wine glass under external pushes.

Dataset-Free Compliant Humanoid Loco-Manipulation with Dynamic Online Posture

- Introduced **OCLO**, a humanoid loco-manipulation system that takes only target hand poses as input and generates its own whole-body posture, trained without human motion data.
- Implemented **whole-body compliance**: IK-controlled arms track force-displaced spring-damper references, while the lower body learns to **adapt to external upper-body force and motion**.
- Enabled the Unitree G1 to **stay balanced under end-effector pulls and pushes**, where ablations without this force training mostly failed.

Undergraduate Research Assistant at Lehigh University – Bethlehem, PA Sept 2023 – Feb 2025

Seed Lab, Advised by Prof. Ning Li

Methods for Setting Device Specifications for Analog In-Memory Computing Inference

- Introduced a systematic approach to **setting device specifications** for analog in-memory computing, identifying device parameters that keep DNN accuracy within 1% of floating point.
- Applied **hardware-aware training** with PyTorch and IBM AIHWKit to make LSTM and ResNet-32 models robust to analog noise before mapping them onto simulated devices.
- Evaluated how **PCM device non-idealities** (conductance range, drift, programming and read noise) affect accuracy over time, through large-scale simulations on HPC clusters.

Undergraduate Research Assistant at Lehigh University – Bethlehem, PA Sept 2023 – May 2024

SwarmsLab, Advised by Prof. David Saldaña

- Built **lighter-than-air blimp robots** for the "Defend the Republic" swarm competition, from 3D-printed frames to onboard IMU, ultrasonic, and camera sensing.
- Implemented **PID control** and camera-based blob tracking for the blimps, with ground-station communication for manual and autonomous modes.
- Developed **ROS 2 ground robots** with AprilTag visual servoing, Kalman-filter SLAM, and A*/RRT/PRM planning.

Undergraduate Research Assistant at Lehigh University – Bethlehem, PA Jun 2023 – Sept 2023

DASH Lab, Advised by Prof. Yu Yang

- Developed a **Flutter mobile app** that collects air-quality data from portable Bluetooth sensors in areas without monitoring stations, supporting future ML-based prediction.
- Added real-time location tracking, live chart and map visualization, and cloud storage on MongoDB Atlas.

Academic Service

Conference Reviewer: IEEE International Conference on Robotics and Automation (ICRA) 2027

Skills

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

Robotics Tools: ROS 2, Gazebo, PyBullet, MuJoCo, Isaac Gym, Isaac Lab, RViz, OMPL

Learning & Computation Tools: PyTorch, TensorFlow, JAX, scikit-learn

Embedded Systems: Arduino, Raspberry Pi

Software Development Tools: Git, Docker, Spark, Spring Boot, Android Studio, Node.js

Databases: MySQL, PostgreSQL, MongoDB, JDBC